

QUEBEC CITY JEAN LESAGE, QC, Canada

WMO: 717080

Lat: 46.8036N

Lon: 71.3817W

Elev: 74

StdP: 100.43

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-23.0	-31.7	0.2	-25.3	-29.0	0.3	-22.4	13.3	-7.0	12.1	-9.4	4.3	260	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.8	21.1	27.3	20.2	25.8	19.3	22.7	26.8	21.5	25.4	20.6	24.3	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.0	25.0	20.2	15.0	24.0	19.1	14.0	22.9	67.3	27.0	63.2	25.4	59.6	24.3	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.6	8.5	DB	-29.7	31.6	2.8	1.3	-31.7	32.5	-33.4	33.3	-35.0	34.0	-37.1	34.9	(4)
(5)			WB	-29.9	24.7	2.8	1.5	-31.9	25.8	-33.5	26.6	-35.1	27.5	-37.1	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-11.5	-9.8	-4.0	3.7	11.7	16.9	19.8	18.7	14.2	7.3	0.3	-7.0	(6)	
	DBStd	11.86	7.00	6.26	5.65	4.21	4.03	3.51	2.73	2.81	3.72	3.90	4.79	6.09	(7)	
	HDD10.0	2798	666	554	434	192	26	1	0	0	6	101	291	527	(8)	
	HDD18.3	4963	924	787	692	438	208	68	16	31	132	341	541	785	(9)	
	CDD10.0	1014	0	0	0	3	79	208	303	269	133	18	1	0	(10)	
	CDD18.3	138	0	0	0	0	3	25	60	41	9	0	0	0	(11)	
	CDH23.3	1004	0	0	0	2	60	214	409	256	63	1	0	0	(12)	
	CDH26.7	192	0	0	0	0	11	47	83	42	9	0	0	0	(13)	
(14) Wind	WSAvg	3.8	4.5	4.4	4.3	4.2	3.9	3.4	3.2	3.1	3.2	3.8	4.0	4.1	(14)	
(15) Precipitation	PrecAvg	1266	96	85	91	82	102	114	123	121	127	99	112	117	(15)	
	PrecMax	1648	159	192	150	154	196	182	200	224	218	171	184	221	(16)	
	PrecMin	988	43	6	21	22	32	40	57	33	36	37	50	47	(17)	
	PrecStd	148	35	46	29	29	44	35	38	53	41	34	34	36	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.8	10.2	20.3	28.2	30.3	30.9	29.7	28.0	21.4	14.8	6.0	(19)	
		MCWB	3.9	2.9	5.4	12.3	17.9	21.7	23.4	22.2	21.1	16.1	12.9	5.4	(20)	
	2%	DB	2.4	2.6	6.5	15.8	24.9	28.0	28.8	27.8	25.1	17.9	10.8	3.4	(21)	
		MCWB	1.6	1.1	3.3	8.9	15.9	20.2	21.5	20.7	19.4	13.9	8.6	2.5	(22)	
	5%	DB	0.6	1.1	4.5	12.6	21.9	25.9	27.2	26.2	22.7	15.3	8.5	2.1	(23)	
		MCWB	0.0	0.1	2.0	7.3	14.5	18.7	20.5	19.8	18.2	12.4	6.6	1.3	(24)	
	10%	DB	-1.6	-0.3	3.0	10.1	19.2	23.8	25.7	24.5	20.6	13.2	6.5	0.9	(25)	
		MCWB	-2.4	-1.2	1.0	5.9	13.0	17.7	19.6	19.1	17.0	10.5	5.0	0.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.2	6.8	14.0	20.2	23.2	24.6	23.3	22.4	17.1	13.2	5.4	(27)	
		MCDB	4.3	4.0	8.4	19.1	25.7	28.1	29.3	27.9	26.7	20.1	14.6	5.8	(28)	
	2%	WB	1.7	1.3	3.8	9.8	17.8	21.5	23.0	22.0	20.5	14.7	9.2	2.7	(29)	
		MCDB	2.2	2.2	5.9	14.2	22.5	26.1	27.0	25.7	23.7	17.3	10.2	3.2	(30)	
	5%	WB	0.0	0.2	2.4	8.0	15.6	20.1	21.7	21.0	19.0	12.8	6.8	1.4	(31)	
		MCDB	0.7	1.1	4.0	11.8	19.8	24.3	25.5	24.6	21.9	15.0	8.1	2.0	(32)	
	10%	WB	-2.2	-1.3	1.2	6.3	13.7	18.7	20.7	20.0	17.4	11.0	5.2	0.3	(33)	
		MCDB	-1.6	-0.3	2.7	9.5	18.7	22.5	24.4	23.5	20.1	12.6	6.5	0.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.1	8.4	8.6	10.5	10.4	10.0	10.0	9.7	7.9	6.6	7.1	(35)	
		MCDBR	8.9	7.9	9.1	13.7	15.4	13.7	12.2	12.1	12.2	11.7	9.5	6.2	(36)	
	5% WB	MCWBR	8.9	7.2	6.8	8.8	8.7	7.5	6.6	6.6	7.4	8.2	7.9	6.1	(37)	
		MCWBR	8.8	7.3	8.0	12.1	13.2	11.6	10.8	10.3	11.1	10.6	8.5	6.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.265	0.284	0.325	0.351	0.367	0.403	0.413	0.395	0.370	0.339	0.308	0.292	(40)	
		taud	2.311	2.246	2.218	2.346	2.397	2.290	2.282	2.334	2.398	2.479	2.471	2.280	(41)	
	Ebn at Noon		831	885	894	904	898	862	848	849	840	818	779	739	(42)	
		Edh at Noon	74	98	119	116	114	128	127	116	100	79	65	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.40	2.31	3.52	4.44	5.06	5.52	5.40	4.69	3.69	2.15	1.25	0.94	(44)	
	RadStd		0.20	0.28	0.29	0.38	0.53	0.35	0.53	0.29	0.31	0.18	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page